

ibm3090_vector_sop_v1json

```
{
  "id": "SOP-IBM3090-002",
  "title": "IBM 3090 Vector Facility (VF): Component Diagnostic & Recovery",
  "system": "IBM 3090",
  "component": "Vector Facility (VF) / Vector Processor TCM",
  "author": "SymbioStreams Semantic Bridge",
  "source": "IBM 3090 Maintenance Library & ES/3090 System Recovery",
  "description": "Diagnostic procedure for isolating hardware failures and parity checks specifically within the optional 3090 Vector Facility.",
  "steps": [
    {
      "step": 1,
      "action": "Indicator Analysis",
      "details": "When a vector operation fails, check the system console for the 'Vector Check' or 'Vector Hard Stop' indicators. Examine the MCIC (Machine Check Interruption Code).",
      "indicator": "Check Byte 2, Bit 3 (Vector Facility Error) and Bit 4 (Vector Facility Failure).",
    },
    {
      "step": 2,
      "action": "Vector Status Word (VSW) Audit",
      "details": "Use the Service Processor to read the Vector Status Word. This register contains the status of the vector execution pipelines and the vector register file.",
      "check": "Identify if the error is localized to the arithmetic pipeline, the logic unit, or the mask register."
    },
    {
      "step": 3,
      "action": "Run Vector Facility Functional Tests (VFFT)",
      "details": "Initiate the VFFT suite from the Service Processor console. These tests verify the vector registers, element addresses, and vector instruction timing.",
      "logic": "If VFFT fails, the diagnostic will point to a specific sub-module within the Vector TCM."
    },
    {
      "step": 4,
      "action": "Concurrent Maintenance / TCM Replacement",
      "details": "The 3090 supports concurrent maintenance. If a specific Vector TCM is faulty, it can often be varied offline while the central processor continues scalar operations.",
      "remedy": "Replace the failing Vector TCM module (e.g., Module C-L1 or D-M2) and reload the vector microcode."
    }
  ],
  "diagnostic_flowchart": [
    {
      "condition": "Vector Check Bit ON (Recoverable)",
      "next": "Retry vector instruction. If persistent, log as degrading TCM."
    },
    {
      "condition": "Vector Hard Stop ON (Non-Recoverable)",
      "next": "Vary Vector Facility OFFLINE. Run standalone diagnostics."
    }
  ]
}
```

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    {  
      "condition": "Parity Error in Vector Registers",  
      "next": "Refresh vector storage from memory. If error persists, replace register stack  
module."  
    }  
  ]  
}
```